

PROJECT DOCUMENTATION

Exports to Asian Countries – Data Analysis

1. Project Overview

This project analyses exports to Asian countries using Microsoft Excel and Power BI. The objective is to clean and process the export dataset, analyse export values and quantities, identify country and commodity patterns, and present the results through an interactive Power BI dashboard.

2. Tools Used

- Microsoft Excel – data inspection, cleaning, processing, PivotTables and preliminary analysis.
- Power Query – data cleaning, transformation, missing-value handling, data-type correction and date fields.
- Microsoft Power BI – data modelling, DAX measures, Matrix analysis, interactive visuals and dashboard creation.

3. Dataset

Dataset: Exports to Asian Countries

Source: CSV dataset supplied for the mini project.

Key columns used:

Category	Columns	Purpose
Date & Geography	date, country_name, region_name, subregion_name	Time and geographic analysis
Country Codes	alpha_3_code, country_code, region_code, subregion_code	Identification
Commodity	hs_code, commodity	Commodity analysis
Quantity	unit, value_qt	Quantity and unit analysis
Export Value	value_rs, value_dl	INR and USD export values
Record ID	id	Record identification and duplicate checking

https://drive.google.com/drive/folders/1H5PWeOQUmMUfurlGazpBPbxJqvn2Fiyy?usp=drive_link

4. Steps Followed

4.1 Data Import

- Imported the CSV dataset into Excel Power Query.
- Reviewed the dataset structure and available columns.
- Used Power Query as the main transformation layer for the large dataset.

4.2 Data Cleaning

- Checked and corrected data types for date, text, whole-number and decimal-number fields.
- Applied Trim and Clean to relevant text fields.
- Checked missing values in important fields.
- Missing Unit values were retained as null/unknown rather than replacing them with zero because a missing unit does not mean zero quantity.
- Checked value_qt, value_rs and value_dl for errors and unusual values.
- Checked the id field for duplicate records.
- Reviewed country and commodity fields for inconsistent or unexpected entries.

4.3 Data Processing

- Created Year, Month Number and Month Name fields from date.
- Created a Year-Month field for monthly trend analysis.
- Prepared the cleaned data for Excel and Power BI analysis.

4.4 Excel Analysis

- Created PivotTables for total export value by country.
- Sorted export values from largest to smallest and applied Top 10 filtering.
- Analysed export value by commodity, region, subregion and year.

4.5 Power BI Data Model

- Used Exports_Cleaned as the fact table.
- Created DimDate, DimCountry and DimCommodity dimension tables.
- Created one-to-many relationships from the dimension tables to Exports_Cleaned.
- Used the model for time, country and commodity filtering.

4.6 DAX Measures

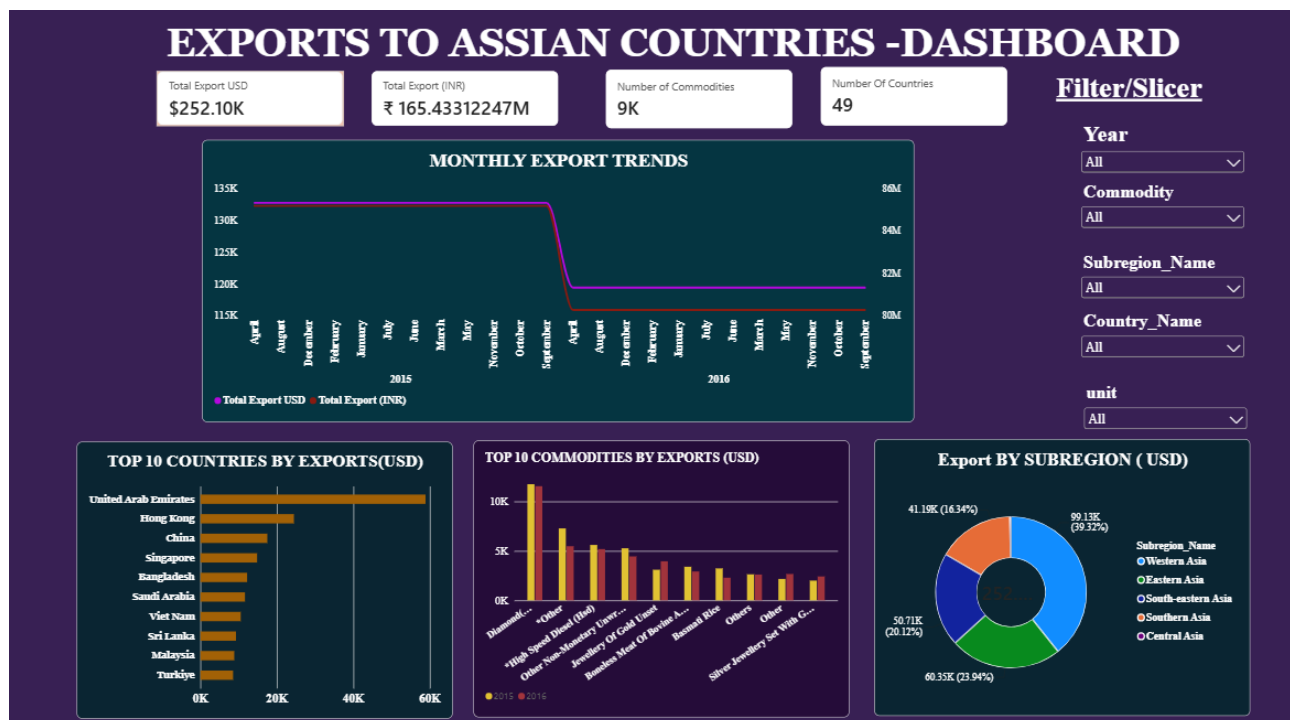
Measure	DAX
Total Export USD	Total Export USD = SUM(Exports_Cleaned[value_dl])
Total Export INR	Total Export INR = SUM(Exports_Cleaned[value_rs])
Total Quantity	Total Quantity = SUM(Exports_Cleaned[value_qt])
Number of Countries	Number of Countries = DISTINCTCOUNT(DimCountry[Country])
Number of Commodities	Number of Commodities = DISTINCTCOUNT(DimCommodity[Commodity])

5. Key Insights

The dashboard is designed to identify leading export destinations, major commodities, time-based trends and geographic patterns. Specific numerical findings should be entered from the final Power BI results.

- Top countries can be ranked by total export value in USD.
- Top commodities can be identified by their contribution to export value.
- Monthly and yearly trends show changes in export activity over time.
- Region and subregion analysis supports geographic comparison.
- Slicers allow interactive analysis by Year, Country, Region, Commodity and Unit.

6. Screenshots



7. Files Included

- Cleaned Excel/CSV file – cleaned and processed export data.
- Power BI .pbix file – data model and interactive dashboard.
- Project Documentation – project report.

8. How to Use

- Open the cleaned Excel/CSV file to review the transformed data and preliminary analysis.
- Open the Power BI .pbix file in Power BI Desktop.
- Use the Year, Country, Region, Commodity and Unit slicers.
- Review KPI cards, charts and the Matrix visual for export analysis.